

In addition to the enzymes reacting with DNA, enzymes which can make DNA on an RNA template is also very useful in gene technology. These enzymes are called reverse transcriptase, because their action is the reverse of the transcription. This is used to make cDNA (copy DNA or complementary DNA) on an mRNA template.

The DNA shearing or cutting with restriction enzymes makes a mixture of DNA fragments of different sizes. In DNA fingerprinting, using PCR, DNA strands having different sizes are obtained. Therefore in many applications with DNA, separation of DNA molecules becomes important. Separation of fragments having different size on a gel matrix is the most practical way of doing this.

Agarose gel electrophoresis: Electrophoresis is a technique that separates large charged molecules, (such as DNA, RNA and proteins) according to their mobility in an electric field. The speed of a molecule moving in an electric field depends on its net charge and the size. In gel electrophoresis, the molecules are moving through small pores of a matrix of gel. This restricts the movement of the molecules and helps in separation according to the size; larger molecules move slowly compared to smaller molecules. In the case of nucleic acids, the net charge depends on the length of the molecule, and therefore the separation is based on the size. For the separation of DNA, Agarose Gel Electrophoresis is the most frequently used technique. Agarose is purified agar, obtained from a type of seaweed. This forms a polysaccharide matrix. In the Agarose gel electrophoresis equipment, the gel is placed in buffer and a cathode and an anode are placed on either end of the gel (Figure 7.29(a)). When a current is applied using a power supply, the negatively charged DNA molecules migrate towards the anode through the gel. When preparing the gel, wells are formed, and the DNA is loaded into these wells. The separated DNA can be stained by ethidium bromide and be visualized by exposing to UV light (Figure 7.29(b)).

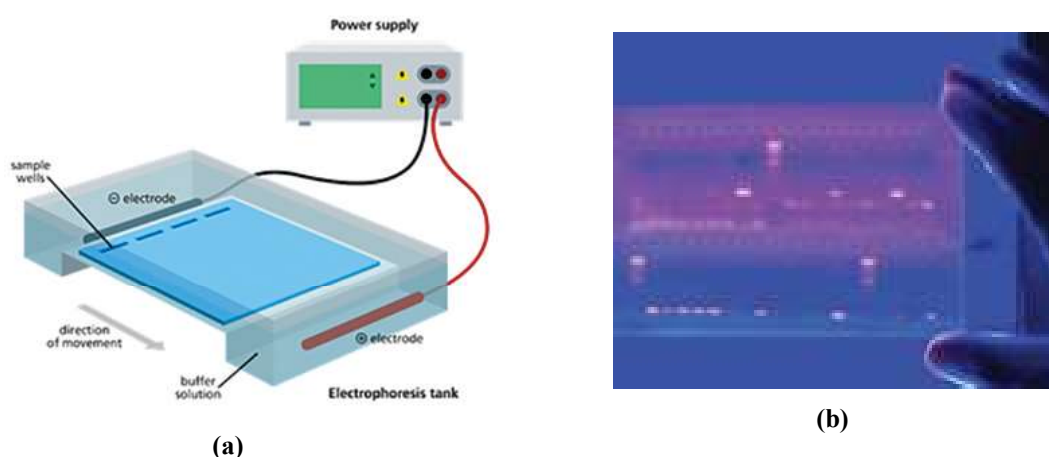


Figure 7.29: (a) Agarose gel electrophoresis equipment (b) Visualizing separated bands of DNA on a gel using UV light

Although stains like ethidium bromide show the presence of double stranded DNA band on an agarose gel, those stains cannot differentiate a band with a specific nucleotide sequence from the others. In order to identify such a band among many other bands, a DNA probe is used.

DNA Probes and hybridization: A DNA probe is a fragment of single stranded labeled DNA used to detect the presence of complementary nucleic acid sequences by hybridization. Labeling is a modification of the DNA strand in such a way that it gives a signal which enables the detection of that DNA strand. The labeling can be done by, for example, incorporation of a radioactive isotope, or addition of a fluorescent molecule to the structure of the probe. This single stranded DNA stretch can hybridize to complementary single strand of DNA or RNA. Therefore before hybridizing with the probe, the double stranded DNA needs to be denatured to make room for the probe. The denatured bands on the gel need to be transferred to a nitrocellulose or nylon filter membrane by a process called Southern Blotting. The bands are then fixed to the membrane. Then the labeled probe is added to the membrane and allowed to renature. The probes will bind strongly only to the complementary sequences, which are fixed to the membrane. When the membrane is washed, the probe is removed except the probes bound to the bands with target nucleotide sequence. If the probe is radioactively labelled, the band with the target sequence can be identified by autoradiography of the membrane. If the probe is labeled with a fluorescent dye, the band can be identified using UV light.

Recombinant DNA Technology

All organisms on Earth have evolved from a common ancestor and their genetic information is stored in DNA, except in some viruses. At the chemical level, DNA is the same in all organisms. Further, all organisms share the same genetic code, so the same peptide is encoded by a gene whether it is expressed in a bacterium, a plant or an animal. This forms the basis of recombinant DNA technology, where DNA from two or more different species are joined together and inserted into a host to obtain a new genetic combination which is of value to science, medicine, agriculture, industry and environmental applications.

Recombinant DNA molecules:
DNA molecules made by laboratory methods of genetic recombination which brings together DNA from different sources, creating sequences that are not found naturally.

All the techniques mentioned below are needed to make a recombinant DNA molecule (rDNA).

They are;

- Isolation of DNA from different sources.
- Restriction digestion of isolated DNA with restriction enzymes.
- Separation of DNA fragments by gel electrophoresis.
- Identification of the correct fragments with desired nucleotide sequences using probes.
- joining the DNA fragments from multiple sources using DNA ligase.

Inserting DNA molecules to host cells is a difficult step. Cells show resistance to DNA uptake. This is important for the survival of organisms because invading DNA can cause genetic alterations which are usually harmful.

Therefore, a large number of copies of the recombinant DNA molecules are necessary to make sure at least a few host cells will receive a copy. If the desired DNA fragment is a shortone, *in vitro* multiplication can be done by a technique called DNA cloning.

DNA Cloning

In DNA cloning, the DNA replication machinery of a host cell is used to copy the desired DNA. However, inserting a fragment of DNA into a host cell will not be copied, unless an origin of replication (Ori) is present in that. Therefore, in order to replicate the recombinant DNA molecule or DNA of interest, it has to be combined with DNA with Ori and that can be replicated independent of the chromosomal DNA (chromosomal DNA replicates only once in a cell division). In a bacterial host, there are several copies of plasmids and also when infected with a bacteriophage, a large number of copies of viral DNA per bacterial cell are present. As such the DNA molecule of interest can be integrated into these self-replication units, which are called vectors.

Vectors: The vectors are vehicles to carry the DNA of interest into a host for multiplication or cloning. The vectors used in cloning of DNA are called cloning vectors. When the vector is carrying the foreign DNA, it is called a recombinant vector.

Making a recombinant vector also follows the same procedure for making the recombinant DNA molecule. Here the gene of interest has to be cut with a restriction enzyme and the vector (plasmid or the viral DNA) should also be cut with the same enzyme. These two should be mixed and allowed to integrate and ligated using DNA ligase (Figure 7.30). The site in the vector, where the DNA to be cloned is inserted is called cloning site. In order to be able to use several restriction enzymes to cut DNA (both the vector and the DNA to be cloned), the cloning site contains a sequence which carry sites for many restriction enzymes. Therefore, this is called multiple cloning site. A host cell, usually a bacterial host which can copy the vector, is then transformed by the recombinant vector. The host then will copy the plasmid carrying the DNA of interest. A number of recombinant plasmid is present in each descendent cell of the colony of bacterial host.

Types of vectors and their differences

Any self-replicating unit in a particular host cell can be used as a vector. In bacteria, plasmids and bacteriophages are used as vectors. Plasmids are present in yeast cells also. Therefore they can also be used as vectors in yeasts. The yeast cloning vectors are called yeast artificial chromosomes or YACs. They are plasmids, but are called chromosomes, because they carry sequences of centromeres. They work as chromosomes when they are linearized. In addition, they also have autonomously replicating sequences (ARS) to enable them to replicate independent of the cell division. All of these vectors carry genes which are not necessary for a vector. These are removed and this space is used to insert the DNA of interest. The yeast vectors should contain sequence of centromere and autonomously replicating sequences (ARS).

As described above, main purpose of cloning vector is to copy DNA in an *in vivo* system. For this, number of copies in a single host has to be high. So this condition is fulfilled by bacterial plasmids, bacteriophages and YACs. The transformation of cells is a very inefficient process. However, using bacteriophages as vectors overcome this problem, because the vector can be inserted to the host cell by the mechanism of infection of the bacteriophage. An advantage of YACs is that they are large, therefore, large amount of DNA can be copied using them. It has also another advantage, because they operate in the eukaryotic system.

Transformation:
Direct uptake of
exogenous DNA from
the surroundings of
a host through its
cell membrane and
incorporation, resulting
in a genetic alteration

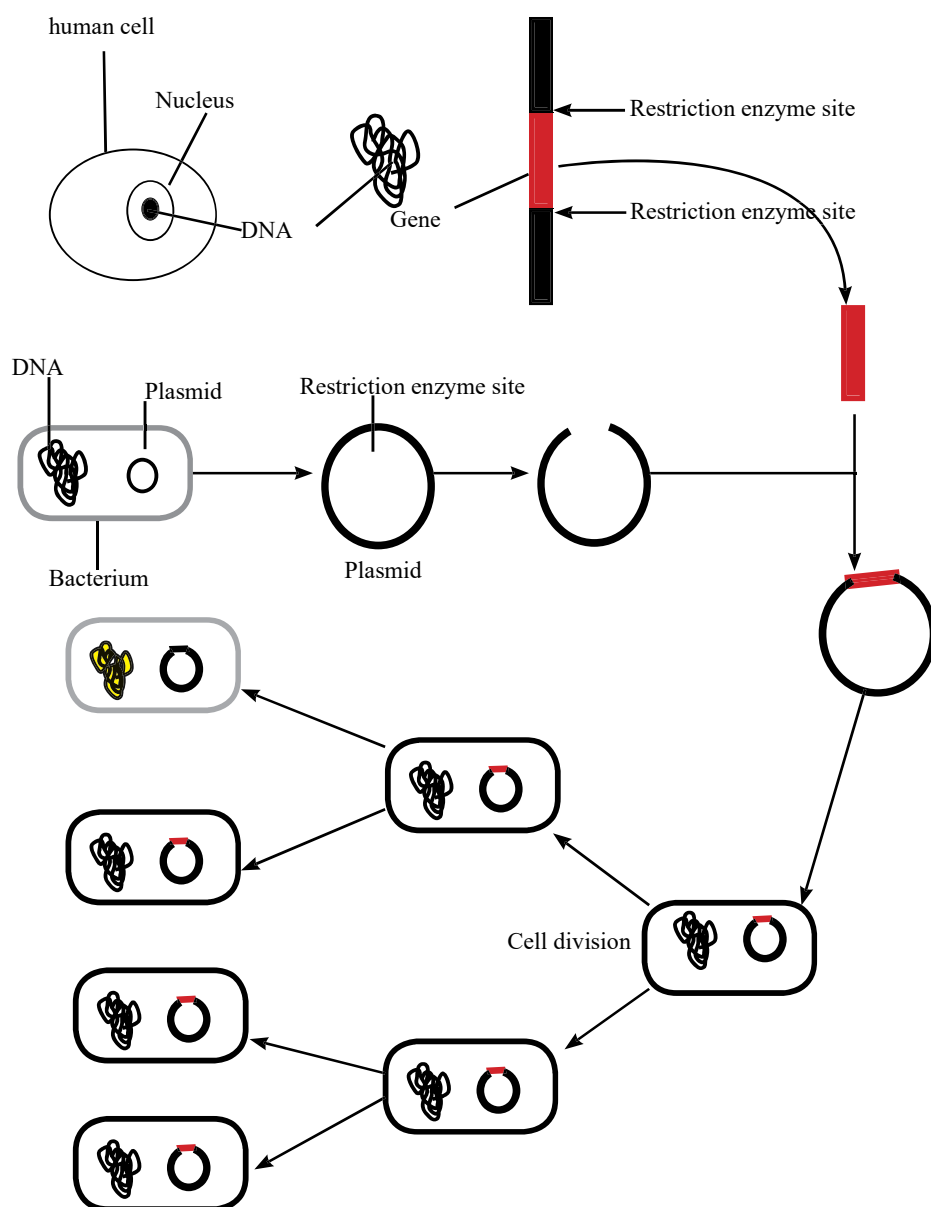


Figure 7.30: Cloning of a gene of interest using a plasmid vector and bacterial host

The copies of the DNA of interest or recombinant DNA is retrieved by harvesting the host cells, releasing the vectors by lysis, isolation of the vectors and cutting the DNA with the same restriction enzymes that was used originally to isolate the fragment of DNA. The recombinant DNA is then isolated after separation and detection on an agarose gel after electrophoresis.

Use of marker genes: The efficiency of the transformation of the host cells with the recombinant plasmid vector is very low. This means, that for one transformed host cell, there are millions or billions of cells that are not transformed. Both transformed and untransformed cells will make

colonies in a suitable medium, but these cannot be distinguished. Therefore, some sort of marker gene should be engineered into the cloning vector, so that the few colonies originating from the transformed cells can be screened among many untransformed cells. Most common markers are antibiotic resistance genes. The host cell is sensitive to the particular antibiotic, and hence will not grow on a medium containing the antibiotic. Since the vector is carrying the gene for resistance, the transformed cells can grow in the medium with antibiotic. Such markers are called selectable markers, since they allow the growth of only the transformed cells.

There is another problem to be solved. The transformation does not necessarily mean that the insert is there. Not all vectors are recombined with the gene of interest. Therefore, another marker is required to distinguish the colonies with the vector containing the insert from those with vector only. The essential features that should be present in a cloning vector are shown in Figure 7.31.

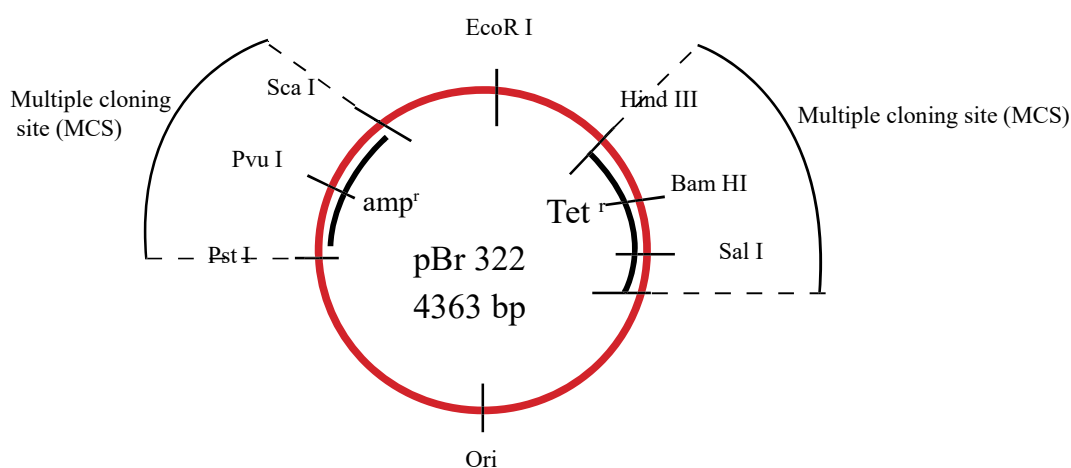


Figure 7.31: An example of a cloning vector (pBR322) showing essential features (ori, multiple cloning site, markers)

DNA Libraries

When a genome is cut into random pieces by shearing using mechanical force or by restriction enzymes, it creates an extremely large number of different sequences, depending on the size of the genome. All these pieces can be integrated into cloning vectors and the recombinant vectors can be used to transform bacterial hosts. These hosts can then be cultivated in a suitable medium for selection of transformed cells and screening for transformed cells with vectors carrying an insert. Since there was no selection for a particular segment of DNA in this case, each transformed cell with an insert may carry a different piece of DNA of the genome selected earlier. When all colonies are isolated and cultivated separately, the collection of these colonies is referred to as a genomic DNA library (Figure 7.32). DNA libraries are, a collection of microbial cultures each propagating a different fragments of a total genomic DNA which are cloned in a population of

identical vectors. Inserts of each colony can be sequenced separately in order to have a complete sequence of the genome. This is how the elucidation of sequence human genome under the Human Genome Project started.

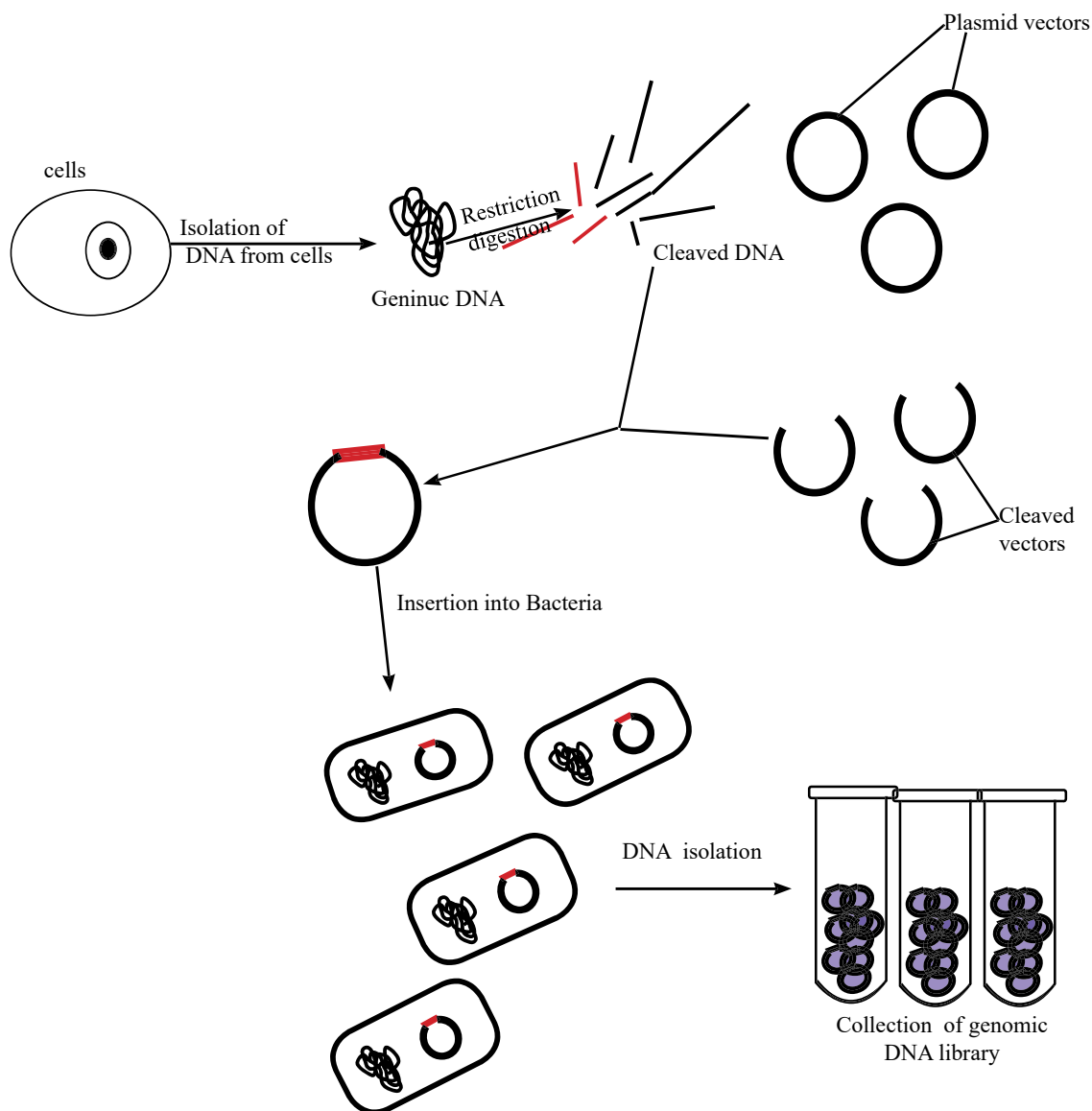


Figure 7.32: Steps in construction of a genomic DNA library

There is another type of DNA libraries. These are called cDNA libraries. These libraries contain complementary DNA obtained by reverse transcription of the mRNAs isolated from cells/ tissues. Collection of mRNAs of a cell is called transcriptome. mRNAs are isolated and reverse transcribed into a DNA strand complementary to the mRNAs. The enzyme used here is reverse transcriptase. The second DNA strand is replicated on the first DNA template using DNA polymerase to obtain double stranded cDNA. The DNA fragments are cloned and similar procedure is followed as for genomic DNA library in order to construct a cDNA library.

DNA libraries are basically used as sources of DNA fragments for sequencing. cDNA libraries also reflect the pattern of expression of genes.

DNA Delivery Systems

A cell contacting foreign DNA is called a transformed cell. Getting the foreign DNA in a cell can be done using several methods.

- **Transformation:** In this method a large number of copies of DNA of interest is mixed (e.g. a recombinant vector) with host cells. This is based on the ability of a cell to take up DNA from its surroundings through the cell membrane. The efficiency is very low in getting the DNA into the cells. Various treatments can increase the competency (ability to take up DNA from outside) of host cells.
- **Transduction:** This method is based on the ability of bacteriophages to infect host cells. Viruses infecting plants and animals can also be used as vectors to deliver foreign DNA into plant and animal hosts. The gene of interest is integrated into the modified viral genome and packaged into the protein capsid. This viral particle can pass the recombinant DNA as in its normal infection process. The capsid protects the DNA and this method is more efficient than transformation.
- **Gene Gun:** In this method, small particles of a heavy metal such as gold are coated with a large number of copies of the DNA of interest and the particles are shot at high velocity into the cells to be transformed. The equipment used is the gene gun (Figure 7.33).



Figure 7.33: Gene Gun

***Agrobacterium* mediated gene transfer:** *Agrobacterium* is a soil bacterium which can infect plants. Their mode of infection is very special. The infection causes a tumour on the plant, and the bacterium is living inside it. The disease is called crown gall disease. The cells in the tumour or gall are actually genetically transformed by a segment of a plasmid of *Agrobacterium*. This plasmid is called Ti (tumour inducing) plasmid (Figure 7.34). A part of this plasmid can actually be transferred to plant genome, and hence called transfer DNA or T-DNA. T-DNA contains genes which induce formation of a tumour and also characters associated with pathogenicity. What is needed for transfer of the DNA is the right and left border sequences of the T-DNA. Therefore

scientists have removed most of the bacterial genes, including virulent genes from T-DNA and have made room for inserting genes of interest into this space between the two border sequences. *Agrobacterium* can deliver this modified T-DNA with inserted gene to plant cells through its infection ability. However plant cells will not become sick because the virulent genes have been removed from T-DNA. This is called disarming the T-DNA.

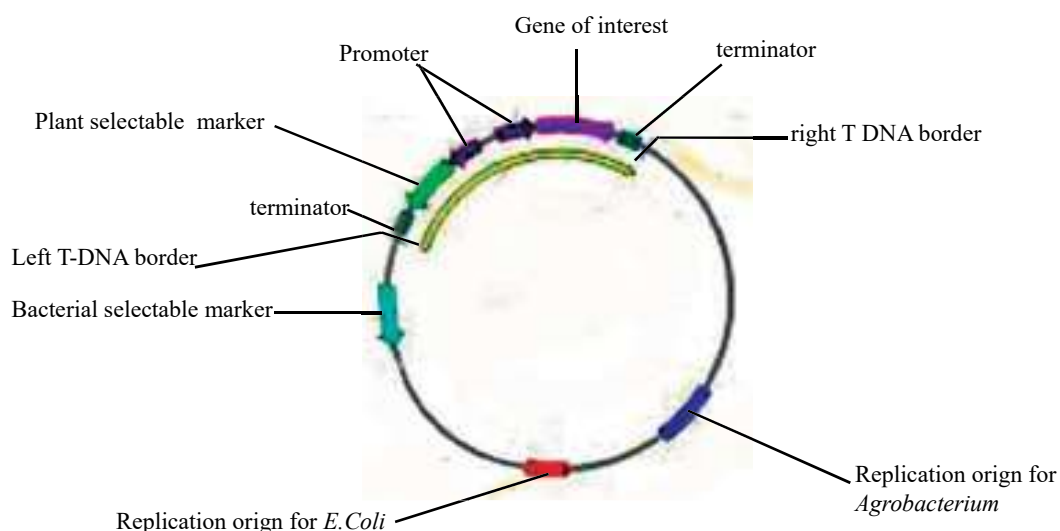


Figure 7.34: Ti Plasmid vector

DNA Analysis

When the morphological characters are used to classify, the number of characters available is limited, so that usually the smallest group identified is a species. When more characters are available, more divisions such as subspecies, strains, varieties etc. become possible. Biochemical properties (i.e. enzyme activities) are useful characters in classification to separate organisms into smaller groups. Since characters are controlled by the combination of the genetics of the organism and its environment, above mentioned characters may vary depending on the environment. Hence if one needs to look how two groups of organisms are similar or different genetically, he has to investigate at the DNA level.

Various techniques for DNA analysis have been developed to facilitate the identification of genetic similarities and differences among organisms and some of these can be used even to identify an individual. These techniques are used in conjunction with techniques such as DNA isolation, gel electrophoresis and use of probes that were discussed above.

Restriction Maps

As shown earlier, restriction enzymes cut dsDNA into fragments at specific sequences. Depending on the number of restriction sites and where they are located, a number of fragments of varying sizes will be generated. Different restriction enzymes cut at different sites and generate different numbers of fragments having different sizes. A restriction map is a diagram showing the position of each restriction site with respect to each other and the distance between these sites (Figure 7.35).

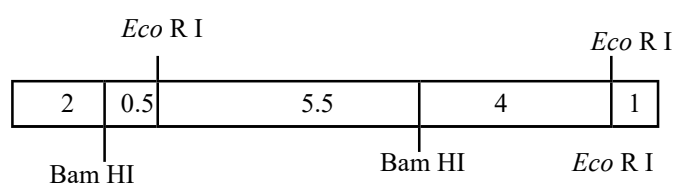


Figure 7.35: Restriction map of a small segment of DNA

Restriction maps are very important in construction of cloning vectors: Cloning vectors are cut by restriction enzymes at the cloning site to insert the fragments of DNA from other sources to the cloning site. Restriction map of a commonly used plasmid vector is shown in Figure 7.36.

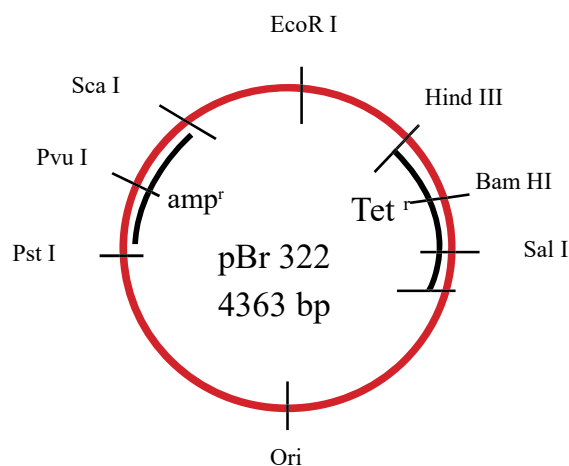


Figure 7.36: Restriction map of a pBR322 plasmid vector

DNA Sequencing

A DNA molecule is made of two complementary and antiparallel strands each comprising four bases; adenine, guanine, cytosine and thymine, arranged linearly in a sequence. DNA sequencing is a process to determine the precise order of these bases in a DNA molecule.

The techniques have been improved greatly since the introduction of DNA sequencing in 1977. The technology of DNA sequencing that was available by the time of obtaining the sequence of entire human genome in 2003, under the human genome project was called first generation sequencing technology. Those methods were time consuming and it was possible to determine the sequence only of short DNA fragments. The technology developed since then to next generation sequencing or second generation sequencing and to the most modern third generation sequencing technology. The most modern technology can sequence strands of lengths of millions of nucleotides, and therefore, the time required for sequencing has reduced greatly. Although the Human Genome Project took 15 years to complete the sequencing of human genome, one can today obtain his/ her genome sequenced in several hours at a cost 1000 US\$ (2018).

With the development of the technology of DNA sequencing, its applications also have been widened.

Applications of DNA sequencing

Molecular Biology: Information on base sequence of DNA is important in understanding the functions of DNA. Genes coding for a polypeptide can be located by studying the DNA sequence. Certain domains in the DNA sequence of a gene specify the function of the protein. For example, whether the protein will be a transmembrane protein or whether it is a DNA binding protein. DNA sequencing has revealed the presence of multiple copies of genes in human genome. Although amino acid sequence of a peptide can be determined using an amino acid sequencer, it has now become easier to work out the amino acid sequence via DNA sequence.

Evolutionary biology: DNA is passed from generation to generation. Over time, changes occur and accumulated in the DNA. Therefore the similarities and differences in the sequences of DNA of individuals within a species and among different species reveal their evolutionary relationships. Sequencing of DNA samples obtained from ancient humans remains preserved, for example, as mummies or buried in ice or as fossilized remains has given insight into when the *Homo sapience* evolved, and how they migrated to conquer the world.

Medicine: Certain families inherit some genetic disorders. Sequencing of DNA reveals if a healthy person is a carrier or not. How a particular disease causing allele is distributed among family members is very important in assessing the risk and to plan a management. Similarly diagnosis of cancer is also possible through DNA sequencing. The response to a drug given for cancer can be followed by sequencing DNA carried in the blood of a patient. If a drug is responding, then the DNA sequences related to cancer must be reduced in blood. DNA isolated from a placenta of a fetus is useful in early diagnosis of presence of genetic disorders.

Forensics: Except for identical twins, it is extremely rare that two individuals share exactly the same sequence of DNA. That means DNA sequencing can identify an individual having the same DNA sequences as DNA material from (blood, hair, sperm, saliva etc) found in a crime scene. Similarly paternity testing is another use of DNA sequencing.

Metagenomics: Microbiome is the totality of microorganisms in a particular habitat including human body and different environments. Conventional methods for studying the microorganisms in a microbiome are based on culturing in pure cultures. However, a large number of microorganisms are unculturable and hence, are largely neglected. Metagenomics is a science in which DNA present in an environment is extracted as community DNA and studying this sample as a whole. Sequencing some specific sequences in this community DNA and analysis using suitable software, will reveal the number of different species and their identity. Some of these are presently known and many others may be new species. Therefore, metagenomics studies are important in ecology, epidemiology and other fields.

DNA Fingerprinting

The unique set of genetic markers of an individual makes its DNA fingerprint or genetic profile. The presence/ absence of markers are nowadays determined by mostly PCR using specific primers for the marker (see below). These markers are called Small Tandem Repeats (STR markers) or Microsatellite DNA. Eukaryotic DNA contains some non-coding sequences where two to six base pairs are repeated tandemly (one after the other) 100 to 1000 times, so that the lengths of these repeats vary. Since they are non-coding, variations do not have an impact on phenotype. These are variable in individuals, so that can be used as makers. The advantages of using STR markers are

- they occur frequently in genome
- easily amplified by PCR
- highly variable polymorphisms and
- a large number of characterized STRs are available.

The method used earlier was probing specific sequences using labeled markers as describes earlier (see DNA Probes and hybridization).

In DNA profiling, a set of markers (probes or PCR primers) is used. A DNA fingerprint cannot be obtained by using one marker, since there are many individuals with the same banding pattern. When more and more markers are used in combination, the probability of finding the same pattern reduced. It has been calculated that, if 13 markers are used the probability becomes between 10 billion to several trillions. Since the world population is in the region of seven billion, it is highly unlikely that two individuals to have the same genetic profile/ fingerprint.

Applications of DNA Fingerprinting

Criminal identification and victim identification (Figure 7.37) – The fingerprints of the suspects are matched with fingerprints obtained from biological material obtained from a crime scene. The expert's opinion on the identity of the criminal is accepted by the Courts of law.

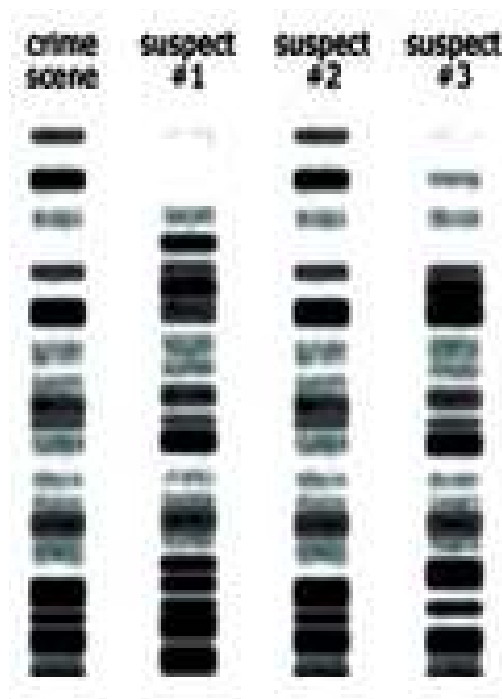


Figure 7.37: Comparison of DNA fingerprints of a sample from the crime scene and three suspects.

The profile of suspect #2 matches with that of the crime scene sample.

Paternity testing (Figure 7.38). The DNA fingerprint of a child is never identical to that of either the father or the mother. However, the child shares some markers from the father and others from the mother. Therefore, when the paternity of a child is in question, DNA profiling can be accurately used to confirm or rule out a person as the father of that particular child (Figure 7.38).

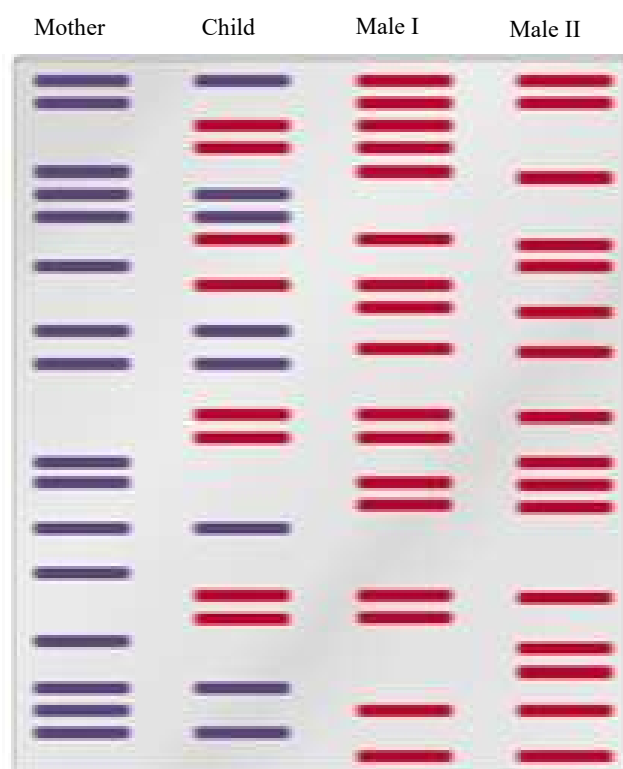


Figure 7.38: Use of DNA fingerprinting for paternity testing

Identifying infectious agents: When probes or primers are known to obtain the fingerprint of a pathogenic infectious organism, DNA fingerprinting can detect the presence or absence of this organism in the patient or in food or water.

Polymerase Chain Reaction (PCR)

Polymerase Chain Reaction (PCR) is used to copy sequences of DNA *in vitro*, mimicking the DNA replication. As in replication, a DNA polymerase enzyme is needed to catalyze the reaction of elongation of the new DNA strand. The raw materials needed are the deoxyribonucleotide triphosphate (dNTPs), which are the four deoxyribonucleotides (dATP, dGTP, dTTP and dCTP). A single stranded template DNA is required. Since the DNA polymerase cannot initiate DNA replication, a primer is also needed. In PCR, a primer is a specific sequence of DNA with a small number of nucleotides (oligonucleotide) which is complementary to a sequence at 3' end of the target DNA to be copied. To copy both the strands two primers, each binding to either of the two strands at 3' end, are used. In the cell, the primer is an RNA sequence. In addition to these, Mg^{2+} is also required. These become the ingredients in PCR mix.

The DNA fragment in which the sequence to be copied is present in dsDNA, and hence it needs to be denatured. Denaturation is done by heating the PCR mix to 95 °C. At this temperature most of the enzymes will denature, and therefore it may be necessary to add DNA polymerase after the denaturation. However, the enzymes of the thermophilic organisms are resistant to high temperature. Therefore, the common heat resistant DNA polymerase used in PCR is Taq DNA polymerase which is obtained from thermophilic bacterium *Thermus aquaticus*. The primer will bind to the complementary sequence of the denatured template DNA. This happens at a lower temperature and this step is called annealing. The annealing temperature will depend on the sequence and the length of the primer. Once the annealing of the primer is achieved, then the primer extension (DNA synthesis) will take place at a different temperature. This is the optimum temperature of the DNA polymerase used. When sufficient time is given, a complementary copy of the template DNA is completed. At the end of this first thermal cycle (denaturation, annealing and extension temperatures), one copy of each strand is produced. However, these are longer than the intended copy of the target DNA sequence (Figure 7.39). After a couple of PCR cycle, an exact copy of the target DNA is synthesized. After this, copies of the target DNA will be produced in an exponential manner (i.e. 2, 4, 8, 16, etc.) after each cycle. A typical PCR will have 35 to 40 cycles. At the end, millions of copies of the desired DNA sequence will be produced from a single template DNA molecule. The PCR product formation in the first three cycles is shown in Figure 7.39.

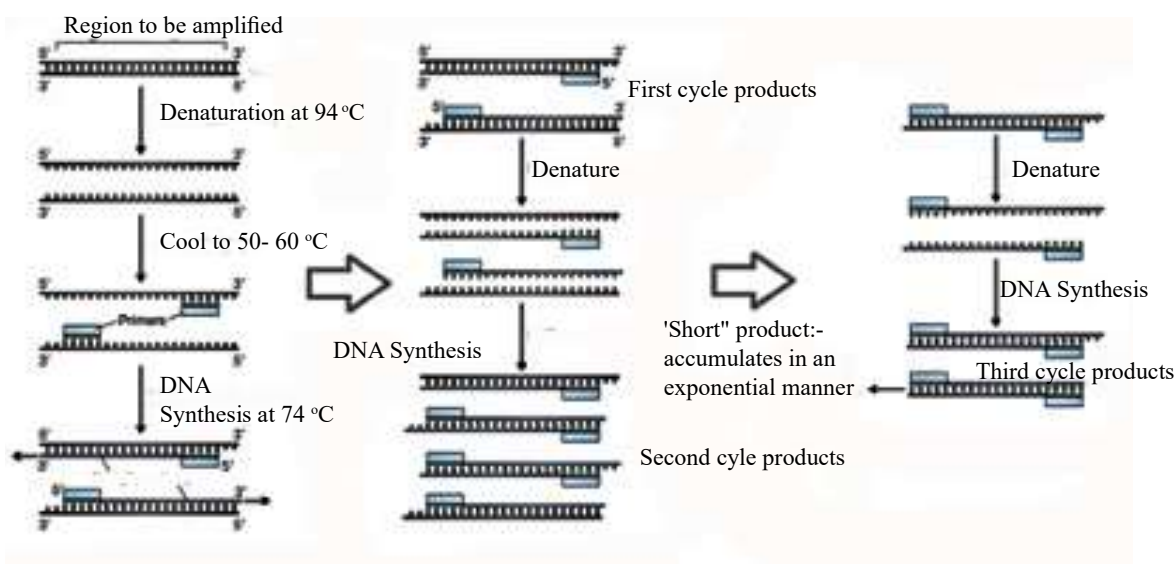


Figure 7.39: PCR products formed at the end of first to third cycles

The repeating cycles are automated and are done in a PCR machine (thermal cycler) (Figure 7.40). PCR mixture is prepared in PCR tubes and these are inserted in to the wells in PCR machine.



Figure 7.40: A PCR machine showing the wells where the PCR tubes are inserted

PCR is a rapid method of obtaining a large number of copies of DNA with high accuracy.

PCR is an essential technique in cloning procedure which allows generation of large amounts of pure DNA from tiny amount of template strand and further study of a particular gene.

Applications of PCR

- Analyzing clinical specimens for the presence of infectious agents (e.g. HIV, hepatitis, malaria)
- Analysis of mutations causing genetic diseases. (e.g. cystic fibrosis, sickle cell anaemia, phenylketonuria)
- Used in forensics laboratories. Since PCR can make a large number of copies from a small number of template DNA, it is especially useful because only a tiny amount of original DNA is required (e.g. droplet of blood or a single hair).
- PCR is an essential technique in cloning procedure which allows generation of large amounts of pure DNA from tiny amount of template strand and further study of a particular gene.
- Sequencing of DNA depends on PCR.
- PCR has been used to identify, and to explore relationships among species in the field of evolutionary biology.
- In anthropology, it is also used to understand the ancient human migration patterns. In archaeology, it has been used to spot the ancient human race.
- PCR commonly used by paleontologists to amplify DNA from extinct species or cryopreserved fossils of millions of years old, and can thus be further studied to elucidate on their evolutionary relationships.

Applications of Genetically Modified Organisms (GMOs)

The host with the extra trait introduced by genetic engineering techniques is called a genetically modified organism (GMO). Some related terms are Genetically Engineered Microorganisms (GEMs), transgenic organisms and living modified organisms (LMOs). The food and feed obtained from GMOs are called Genetically Modified Food (GMF). Since majority of the crop plants, farm animals and pets animals used today are genetically modified by domestication, it is important to note that what is referred to as GMOs in today's context are essentially the organisms derived as a result of rDNA technology.

(Note: It is important to realize that animal cloning is not a gene technology. It does not include any of the above mentioned steps.)

The steps in the process making a genetically modified plant or an animal are as follows.

1. Identification of a suitable gene
2. Isolation and purification of the gene
3. Amplification of the gene by cloning
4. *In vitro* modifications of the gene of interest
5. Amplification of modified gene by cloning
6. Transformation of the recipient cells (microbial cells, cells or protoplasts of plants or fertilized eggs of animals)
7. Screening for expression of the inserted gene
8. Monitoring for stable integration of the modified gene
9. Backcrossing to introduce the new trait to other varieties of crops and breeds of animals

The applications of gene technology can be found in many fields, including agriculture, medicine and industry.

Applications of GMO in Agriculture

With the growing human population and decrease in agriculture land, it becomes obvious that , the crop yield per unit area needs to be increased. In order to achieve an economically sustainable agriculture, a higher crop yield should be obtained with a low cost of production. Improving the quality of foods is also a major concern in agriculture, in addition to the quantity. The green revolution from 1930s to 1960s increased the crop yield by introducing high yielding crops and use of artificial fertilizer and pesticides. However, impact of green revolution is also limited and is now being overcome by the genetically modified plants (GM crops). When a single plant cell is genetically modified, it can be regenerated in a plant, as plant cells are totipotent. When a useful

trait is introduced into a crop, it can be introduced to other varieties of the same crop by plant breeding.

Most significant contributions in crop yield enhancement by gene technology in agriculture are due to the production of GM crops that are resistant to

- pests and diseases
- herbicides and
- environmental stresses.

In addition, crops with higher nutritional values are also available, e.g. golden rice enriched with vitamin A, increased triglycerides content in Canola oil.

Plants Resistant to Pests: Several GM crops have been engineered with a gene producing a toxic protein that kills the plant eating larval stages of certain lepidopteron and coleopteron insects. Most widely cultivated pest resistant GM plants are cotton, maize, canola and potato. A GM rice variety resistant to lepidopteron insects is also available.

The protein is called Bt toxin, since the protein originally came from a bacterium; *Bacillus thuringiensis*. Different strains of this bacterium produce several different Bt toxins. When larvae feed on plant parts that express Bt toxins, they die due to the ingestion of the toxin.

Bt toxins are not harmful to mammals and are therefore considered safe for human consumption. However Bt maize (Bt corn) is grown mostly for biofuel and for animal feed. Since the toxin is present in the plant tissues, the only insects that are killed are plant pests and therefore Bt crops are also considered safe for beneficial insects. Figure 7.41 show some insect pests of corn and cotton. Bt toxins are natural products and therefore, are biodegradable.

However, the insects, when exposed to the same toxin over the time, develop resistance to that toxin, rendering the GM crop useless. Several solutions to delay the resistance development in insects have been proposed. Since the pollen grains containing the toxin can escape from the field where Bt crop is planted, insects that are not feeding on the crop may also be killed by accidental ingestion of such pollen. Therefore, Bt crops has a potential danger for non-target insects.

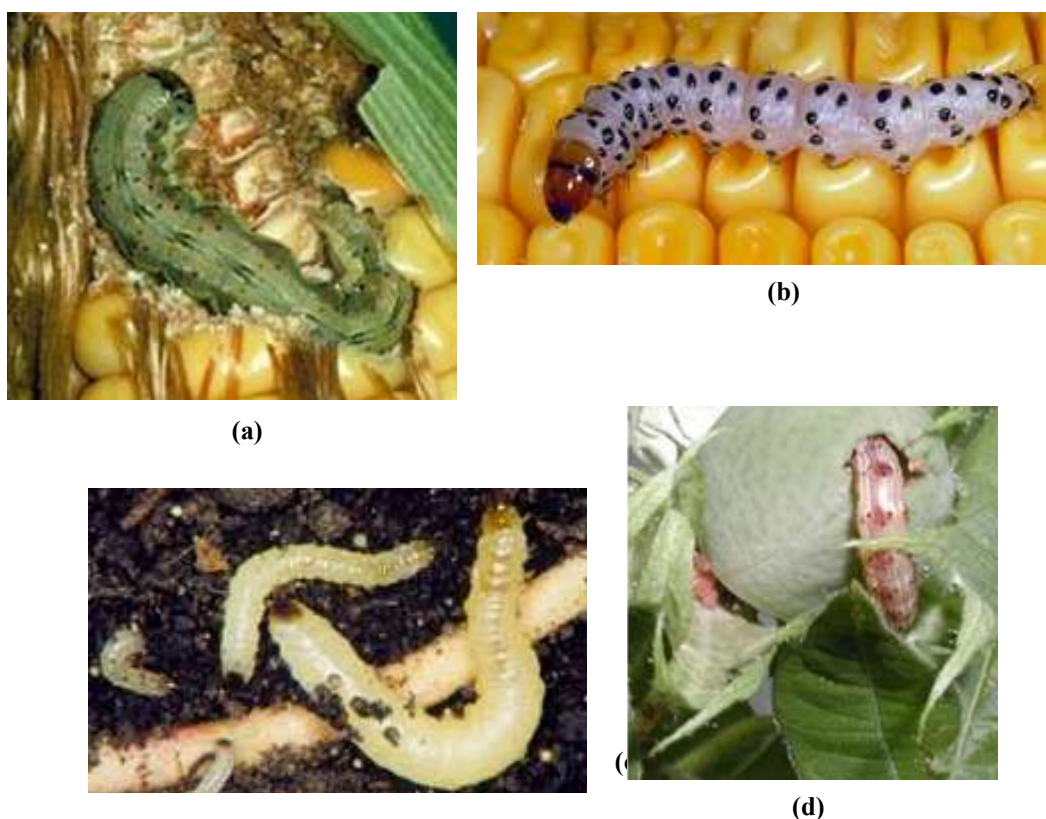


Figure 7.41: Some insect pests of maize: a) Corn earworm, b) European corn borer and c) Corn rootworm. (d) Cotton bollworm

Plants Resistant to Diseases

A well known example of disease resistant crops developed by genetic engineering is the availability of new varieties of papaya resistant to papaya ringspot virus (PRSV). This virus is limiting the success of papaya cultivation worldwide. The same virus also attacks the cucurbits. Squash with resistance to the virus has been successfully developed and cultivated.

Other examples for disease resistant crops are potato varieties resistant to Potato Virus Y (PVY), Potato Leaf Roll Virus (PLRV) and late blight disease.



Figure 7.43: Infected papaya fruit with papaya ring spot virus

Plants Resistant to Herbicides: The herbicide tolerant crops (HTCs) are developed in order to be able to spray a broadspectrum herbicide to control weeds when the crop has been established in the field. When the crop is resistant to the herbicide that is sprayed, it can kill all the weeds without harming the crop. The advantage in this is that the farmers can wait and see if the weeds become a problem and can use herbicide only when it is necessary. This reduces the herbicide usage. However if the same herbicide is used repeatedly, the weeds resistant to that particular herbicide will develop. These are called superweeds. When a crop with resistance to one herbicide is planted after a GM crop with resistance to the same herbicide, leftover seeds of the previous crop may germinate and become a weed which cannot be controlled by the same herbicide. In order to avoid these problems, rotation of crops with tolerance to different herbicides can be practiced.

The best known examples of herbicide tolerant crops (HTCs) are crops modified to tolerate the herbicide glyphosate. These plants are known as “RoundUP Ready” crops because the commercial name of glyphosate is RoundUP.. The examples of commercially available RoundUP Ready crops are cotton, maize, canola, soybean, sugar beet and wheat.

Another type of popular herbicide tolerant crops are called “Liberty Link” and “InVigor” and are resistant to glufosinate. Examples of glufosinate resistant crops include cotton, maize, canola, soybean, sugar beet and rice.

Cotton modified to tolerate Bromoxinol are called BXN Cotton.

GM Plants with other Important Traits

Other agriculturally important traits include improving product quality.

One of the priorities in this area is to increase the nutritional value of crops.

- GM canola varieties with increased triglyceride component and with increased phytase enzyme to breakdown indigestible plant phytate to release phosphorus, GM potato with reduced amylase and increased amylopectin content and soybean with increased oleic acid content in the seed are commercially available.
- A rice variety named Golden rice with increased levels of provitamin A is an interesting example of product quality enhancement. The commercially available variety is modified with genes from *Pantoea ananatis*, a plant pathogenic bacterium containing a yellow pigment.
- Tomato with delayed fruit ripening to increase the flavour and reduce the rate of softening is another interesting development in GM crops. The source of the gene is tomato itself. A part of the gene is copied in reverse direction by changing the orientation of the promoter.
- Non-browning apples due to reduced polyphenol oxidation and maize with increased thermostability of amylase which helps in bioethanol production are other examples.
-

Among plants that can be genetically modified to tolerate the environmental stresses, only maize and soybean having drought resistance have been commercialized.

Applications in Medicine

Human insulin, vaccines and other therapeutics are being produced using GMOs. These pharmaceuticals produced by GMO are cheaper, since mass production at low cost is possible and are considered safer.

Producing human insulin by genetically engineered *E. coli* is the best known example of the use of genetically modified organisms in producing pharmaceuticals. Insulin extracted from animals had various side effects on diabetic patients. The cost of production was also very high due to the limited amount of source to extract insulin. Since those insulins are not the same as human insulin, they were less effective. Nowadays the entire insulin supply is from *E.coli* genetically engineered by inserting insulin gene from human. Therefore the bacteria derived insulin is exactly the same as its original product.

Hepatitis B vaccine used today is a recombinant vaccine produced in yeast. Engineering plants to produce vaccines in their edible parts is a concept being experimented. The idea is to express an antigenic protein in the plant cells. When the edible part with the antigen (e.g. fruit) is consumed by someone; the person will develop antibodies against the antigen, so that immunity is developed against a particular disease. These are called edible vaccines, and if this becomes successful, low cost, safe vaccines can be produced and the delivery of the vaccine would be painless. Storage will also not be a main problem. Therefore these are more important in less developed regions of the world.

GM mammalian cells grown in cell culture are used to extract factor VIII used to treat haemophiliacs and tissue plasminogen activator (tPA) used to treat heart attack and stroke patients.

After the completion of human genome project, the causes for various genetic diseases are being identified with ease and speed. When the cause is identified, how to correct the mistake in the defective genes can also be worked out. The incorrect gene then can be replaced with the correct one by gene technology. If the problem is the expression of a specific gene, this technology can also influence the expression of that gene. This treatment is called gene therapy or human gene transfer. The DNA is delivered to the target cells extracted from the patient by using a viral vector or as naked DNA using various techniques. The cells with corrected gene are reintroduced into the appropriate tissue of the patient. Although the concept of gene therapy has a long history since 1972, only a few applications are available to date.

A gene therapy to treat leukemia is the first of that type in the USA. Another example of gene therapy is to replace the mutated beta-globin gene causing sickle cell anemia with the correct gene. During this procedure, bone marrow derived haematopoietic stem cells are extracted from the patient, the normal beta globin gene is inserted into these cells and return the modified cells back into the patient. The corrected bone marrow stem cells will produce normal erythrocytes.

The concept of personalized medicine is being developed, which is based on genetic information of the patient to treat or prevent diseases.

GM insects have been employed to control insect vector born disease. GM mosquitoes have been engineered which do not allow entry of malaria parasite into the gut, therefore breaking the life cycle of the parasite. Release of these mosquitoes into the wild would reduce incidence of malaria. Another example is the making of GM male mosquitoes carrying a male sterile gene. Mass release of these male sterile insects results in mating with females, but not producing offspring. This technology is call ‘sterile insect technology’ (SIT). In field trials in Brazil, the *Aedes aegypti* populations have reduced by 95% by the introduction of sterile GM males.

Applications in Industry

Use of GMOs in industry has made it possible to produce novel products with low cost while minimizing environmental impacts. Industries based on organisms or their products run at ambient temperatures and pressures, demanding less energy.

Apart from the booming pharmaceutical industry based on GMOs and GM crop industry, some products originated from GMOs are produced industrially. Some enzymes for food processing and detergents are products of GEMs. Chymosin (rennin or rennet) is the first approved enzyme produced by GMO. This is used in cheese industry to coagulate milk to separate whey. Chymosin has been obtained by extracting it from stomachs of slaughtered calves. Since the supply was limited, the cost was high and it affected dairy industry. The chymosin gene from cattle has been engineered in yeast cells and these recombinant yeasts are the source of chymosin today. The price has gone down considerably and the product is purer and devoid of contaminants of animal origin. Amylomaltase is another enzyme produced by a GM *Bacillus* sp. This enzyme modifies starch to be used as an ingredient in dairy industry.

Aspartame, a strong sweetener, is a food additives produced by GM *E. coli*.

Concerns of using GMOs

The most significant risk factor in use of GMOs is their possible unintended effects. Since this is a relatively new technology, public is reluctant to accept it readily. However the public also accepts the unlimited potentials of the GMOs. The debate between pro-GMO and anti-GMO groups, organizations and individuals is intensive. Some of the controversial issues about GMOs are mentioned below.

Health Issues

Health Issues

1. Data of some experiments involving rats, mice and other animals showed some health implications after feeding with genetically modified food such as potato, maize, tomato and soybean. The reports include damage to various tissues such as stomach, liver and kidney and even increase in deaths. However, many other scientists question the methodology used in such experiments and they also claim that they could not reproduce the results. Therefore, more independent research is needed to confirm or to reject such claims.
2. Allergy development due to consumption of GM food or inhaling pollen of GM crops is discussed as another health issue. The integration of foreign DNA in the host cells may change gene expression or cause mutations, leading to unpredictable products. Some of them may be allergenic, toxic or even carcinogenic. However, concrete scientific research findings are not available or the findings are doubtful. The technology is now developed so that the insertion can be done at precise locations without disturbing other functions of the host.
3. The possibility of horizontal gene transfer of antibiotic resistant genes used as marker gene is also highlighted as a potential health issue. The GM foods contain such genes and are ingested in large quantities by consumers. Although this is a possibility, all organisms have barriers to horizontal gene transfer and therefore, chance of horizontal gene transfer into humans is very low. Horizontal gene transfer among bacteria is more probable, and hence transfer of antibiotic resistance to pathogenic organisms may have some health related issues. However, the antibiotics used in rDNA technology are not used in chemotherapy.

On the other hand, the humans as well as all other animals have been eating food of animal or plant origin ever since they were evolved, but there is no evidence to show genes transferred due to eating food.

Environmental Issues

1. The development of insect tolerant crops may harm the non-target insect by accidentally ingesting the toxin produced in the GM crops. The toxin may be dispersed in pollen and deposit on non-crop plants which the insects feed on. An experiment has shown that Monarch butterfly larvae feeding on milk weed leaves dusted with pollen from GM crops die. However, proponents of GM argue that the amount of pollen dusted on the leaves is very much higher than the amount that could be deposited naturally.
2. Cross pollination will transfer the transgene (the foreign gene) to other non-GM varieties of the same crop as well as wild relatives of the crop. It can therefore, contaminate organic or non-GM farming.
3. When Bt gene is transferred to wild plants, the insects feeding on them will die causing environmental imbalances.
4. When the herbicide resistance genes are transferred to weeds, they cannot be controlled using the same herbicide, and therefore become super weeds.
5. The spread of a foreign gene in naturally growing plants is called gene pollution.
6. Since the herbicide tolerant crops are resistant to a particular weedicide, farmers may tend to over use this herbicide to keep their fields clean. If that is the case herbicide tolerant weeds will develop by frequent exposure of the weeds to the same herbicide. It is however argued that the farmers will spend money just to have a clean field, and they can use a "wait and see approach" in order to spray herbicides only when necessary rather than spraying as a precaution. Rotation of crops with tolerance to different herbicides will avoid development of such super weeds.
7. As GM crops are being accepted by farmers as well as consumers, the extent of land under cultivation will be dominated by GM crops limited to very few varieties. When the crop diversity is reduced to such a small number, the tolerance to environmental impacts become very low and a single environmental event may wipe out entire crop fields, leading to food scarcity.
8. Narrowing down of crop diversity will contribute to loss of genes from the crop gene pool.

Socio-economic Issues

1. The newly developed GM crop varieties will be owned by the developers and patented. Therefore the farmers are compelled to purchase their seeds every year from giant seed companies with a monopoly, spending a lot of money. Poor farmers may not be able afford to buy the seeds, and therefore, there is a risk of widening the gap between the rich and the poor farmers.

2. There is a growing public concern whether it is ethically correct to patent crops and biological resources including genes that are found in nature. Some crops and products that have been developed traditionally and used by indigenous people have been patented under some biotechnology companies.
3. The consumer has the right to decide whether he/ she purchases GM or non-GM food. To secure this right, the regulatory agencies need to implement a system of labeling to indicate clearly whether the product is GM or not and if it is GM, then what are the changes that have been made. Labeling is mandatory in some countries. However products labeled non-GM have often been found to be contaminated with GM when tested.
4. Biological resources of a country or region with high biodiversity and traditional knowledge are taken away by biotechnology companies without any authorization by the countries and people or without paying compensation for product development. This is called biopiracy.
5. The manipulation of the nature in making GMOs contradicts the beliefs of certain religions.

In order to address potential risks and hazards of GMO, GMF and associated processes, a thorough testing and screening procedures have come into effect. As such the process of producing and commercialization of GMO and GMF are under strict control of several legislations and authorities in order to safeguard the consumer, society and environment. Approval of some GMOs has taken 25 years from the creation to market (e.g. GM Atlantic Salmon which grow twice as faster than non GM).

An example of an international agreement is Cartagena Protocol. Many countries also have their own legislations, such as National Biosafety Framework of Sri Lanka.

Cartagena Protocol on Biosafety

Cartagena Protocol on Biosafety to the Convention on Biological Diversity is an international agreement signed on 15th May, of 2000 in Montreal, Canada as a supplement to the Convention on Biological Diversity (CBD) signed in 1992, at the Rio- Earth Summit and is effective from 1993. Cartagena Protocol on Biosafety came into effect from 11th September, of 2003. It is called Cartagena Protocol on Biosafety since it had been originally scheduled to sign in Cartagena in Colombia. The provisions in CBD were not sufficient to cover many aspects in biodiversity related to the GMOs. The number of signatories to the Cartagena Protocol has exceeded 100. This includes Sri Lanka which gave formal consent to the protocol on 28th April, of 2004.

The aim of the Cartagena Protocol on Biosafety is to protect the biological diversity from the

potential risks caused by genetically modified organisms or living modified organisms (LMOs) created as a result of modern biotechnology. CBD defined biotechnology as “any technology that uses biological systems, living organisms, or derivatives thereof, to make or modify products or processes for specific use”. The Protocol is based on ‘precautionary principle’ of CBD so that products of new biotechnologies should follow rigorous control measures to avoid any potential risk to the environment or human health and is applied to transboundary movement, transit, handling and use of LMOs that may have adverse effect on the conservation and sustainable use of biological diversity. This also applies to risks to human health. The provisions of the Protocol are intended to allow developing nations to balance public health against economic benefits. Countries and states may take appropriate actions to limit entering of an LMO to their territory if they feel that there is lack of scientific information to ensure that the LMO is safe on environment or human health. The LMOs maybe intended to be introduced into the environment or to be used as food or feed. The shipment should accompany with appropriate documentation which identify the LMO and whom to contact for further information. Sufficient information should be supplied by the importer or exporter to the importing party to arrive at an informed decision on accepting or rejecting the LMO imports and how to handle them in a safe manner.

The Protocol established a ‘Biosafety Clearing House’ (BCH) to assist parties to implement the protocol by facilitation exchange of scientific, technical environmental and legal information on and experience with movement of LMOs.

Sri Lanka signed the Protocol in May, 2000 and it is in force in Sri Lanka from July, 2004. Ministry of Environment and Natural Resources was identified as the responsible institution for coordinating the activities related to the Protocol.

National Biosafety Framework of Sri Lanka

The drafting of the National Biosafety Framework of Sri Lanka (NBFSL) was completed in 2005 by Ministry of Environment and Natural Resources (now Ministry of Mahaweli Development and Environment, MoMDE). This is based on a precautionary approach in accordance with Cartagena Protocol on Biosafety with the objective of ensuring that the risks likely to be caused by modern biotechnology and its products will be minimized and biodiversity, human health and environment will be protected in a maximum way regulating the transboundary movements through formulation of relevant policies, regulations, technical guidelines and establishment of management bodies and supervisory mechanisms.

National Biosafety Framework of Sri Lanka is the starting point for a permanent legislative framework on Biosafety in Sri Lanka. Based on the NBFSL two policies have been formulated.

- National Biosafety, Policy 2005 is one of those. The policy on biosafety sets the overall framework in which adequate safety measures will be developed and put into force to minimize possible risks to human health and the environment while extracting maximum benefits from any potential that modern biotechnology may offer.
- A National Policy on “Access to Biological Resources, sustainable Use and Benefit Sharing” has been formulated in 2013 by MoMDE with the goal of “ensuring conservation and sustainable use of biological resources and the fair and equitable sharing of benefits arising from them” in accordance with Cartagena Protocol on Biosafety and the National Biosafety Framework of Sri Lanka.

However these policies have not yet been enacted by law.

The Target 12 of the National Biodiversity Strategic Action Plan for 2016-2022 of Biodiversity Secretariat of the MoMDE states that “by 2022 biosafety is assured”. The actions to be taken in this direction are

- (1) strengthening policy on biosafety
- (2) implementing biosafety master plan and formulate biosafety legislature
- (3) strengthening risk assessment procedures for new technologies
- (4) strengthening capacity for risk assessment
- (5) developing and implementing legal instruments to protect native biodiversity and indigenous crops from contamination from GMOs and
- (6) enhancing Sri Lanka’s scientific capacity on biosafety.

References

1. Biology, A Global Approach, 10th edition, Campbell Reece, Urry, Cain, Wasserman. Minorsky, Jackson.
2. NCBI

08

Environmental Biology

What is environment ? The environment is the complex of physical, chemical, and biotic factors that act upon an organism or an ecological community and ultimately determine its form and survival

What is environmental biology?

Definition: A scientific study of the origins, functions, relationships, interactions, and natural history of living populations, communities, species, and ecosystems in relation to dynamic environmental processes.

In other words, Environmental biology is the study of different biological components of the environment and its relationship is with each other and with the abiotic components.

Why learn Environmental Biology?

By learning of environmental biology will be able to understand the structure and the functions of the environment we live in. Specifically to:

- understand how biotic and abiotic components of the environment are linked and interact with each other,
- recognize organizational levels of the environment,
- learn about major processes of an ecosystem,
- recognize different components of terrestrial and aquatic ecology,
- understand how human activities are altering ecological systems and
- understand how natural changes in the environment (such as climate change, flooding, erosion etc) affect species and habitats.

Organizational levels of the environment

Several levels of organization are recognized by environmental biologists.

Levels of organization include the individual/ organism , population, community, ecosystem, and

biosphere.

- Individual/ Organism – The first level is an individual. An individual is any organism or living thing. An organism has its own characters in physiology, evolution and behavior in relation to environmental factors.
- Population –A Group of individuals of the same species, living in the same area and producing fertile offspring through interbreeding.
- Community – A collection of populations of different species living in the same area interacting with each other.
- Ecosystem – An ecosystem is a collection of communities as well as the abiotic factors with which they interact.
- Biosphere – includes the entire portion of Earth that is inhabited by life.

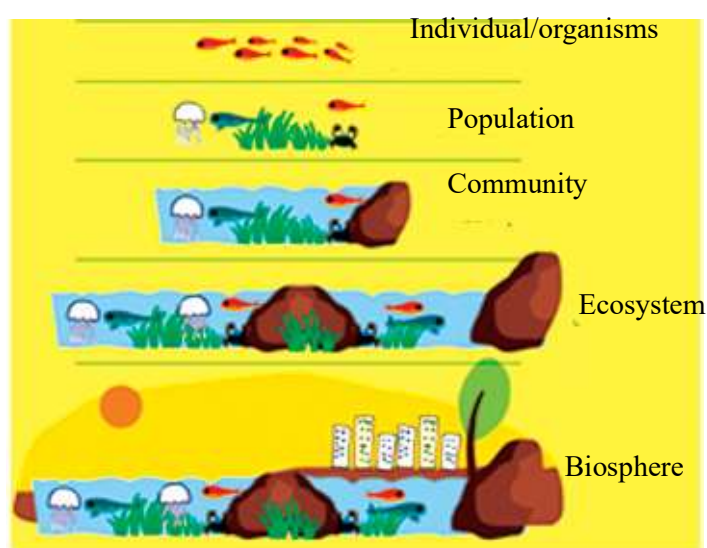


Figure 8.1: Organizational levels of an ecosystem

Major biotic and abiotic components in the environment

All Organisms live in a complex environment that includes several components which are categorized under two major groups:

- Abiotic components – The nonliving components (physical and chemical factors) of the environment such as water, air, light, temperature, nutrients, sunlight, soil. Abiotic resources are usually obtained from the lithosphere, atmosphere, and hydrosphere. Abiotic components limit the geographic range (distribution) and the abundance of species.
- Biotic components – the living components of the ecosystem include plants, animals, fungi, bacteria and protists etc.

Structure and function of ecosystems:

The concept of “Niche” and “habitat”

To understand how an ecosystem work, one should first look into the concepts of “niche” and “habitat.”

Niche

A niche is the role that a particular organism plays in the ecosystem. Each organism plays a particular role in its ecosystem. In other words, a niche is how an organism “makes its living.” This involves how the organism gets its energy, which usually has to do with what an organism eats, and how the organism passes that energy through the ecosystem. An organism's niche also includes how the organism interacts with other organisms, its role in recycling nutrients, tolerance to environment conditions such as temperature, soil moisture, etc. Therefore niche of an organism describes what it needs to live and what it does in a particular ecosystem.

Habitat

The habitat is the physical area where a species lives. Many environment factors are used to describe a habitat. The average amount of sunlight received each day, the range of annual temperatures, and average yearly (annual) rainfall describe a habitat. These and other abiotic factors of the environment such as soil condition will determine the kind of traits an organism must have in order to survive there. A habitat should not be confused with an ecosystem: the habitat is the actual place of the ecosystem, whereas the ecosystem includes both the biotic and abiotic factors of the habitat.

Interactions between abiotic and biotic components of the environment

There are different types of interactions in an ecosystem. They are:

- Biotic-biotic interactions- e.g. competition, feeding relationships, symbiotic relationships between individuals and species.
- Biotic-abiotic interactions- e.g. water uptake of plants from soil
- Abiotic-abiotic interactions-e.g. chemical reactions occurring in the soil.

Feeding relationships within an ecosystem develop food webs and food chains through which energy is passed from one organism to another.

Energy transfer in an ecosystem

The main energy provider for the earth is the sun. Using sunlight the green plants produce their own energy and store in tissues. The source of energy required by all living organisms is the chemical energy of their food. Ultimately, chemical energy in any organism is obtained by the conversion of the radiant energy of sun which are stored in tissues of plants and animals.

Trophic level:

The producers and consumers in ecosystem can be arranged into several feeding groups, each known as trophic level (feeding level). In any ecosystem, producers represent the first trophic level, herbivores represent the second trophic level, primary carnivores represent the third trophic level and top carnivores represent other levels.

Food chain

A food chain is a linear sequence of organisms through which nutrients and energy pass from one trophic level to another trophic level of an ecosystem beginning with a primary producer. Let's look at the parts of a typical food chain, starting from the bottom (the producers) and moving upward.

primary producers

At the base of the food chain lie the primary producers. The **primary producers** are autotrophs and are most often photosynthetic organisms such as plants, algae, or cyanobacteria. These organisms convert light energy to chemical energy or in another words produce organic matter.

Primary production – is the amount of organic matter produced by autotrophs in a given area during a given period of time. (Unit: $\text{gm}^{-2}\text{day}^{-1}$ or $\text{kg ha}^{-1}\text{year}^{-1}$)

The organisms that consume (eat) the primary producers are called primary consumers. Primary consumers are usually herbivores (plant-eaters), though they may be algae eaters or bacteria eaters.

The organisms that consume the primary consumers are called secondary consumers. Secondary consumers are generally carnivores (meat-eaters).

The organisms that consume the secondary consumers are called tertiary consumers. These are carnivore-eating carnivores, such as Eagles or big fish species.

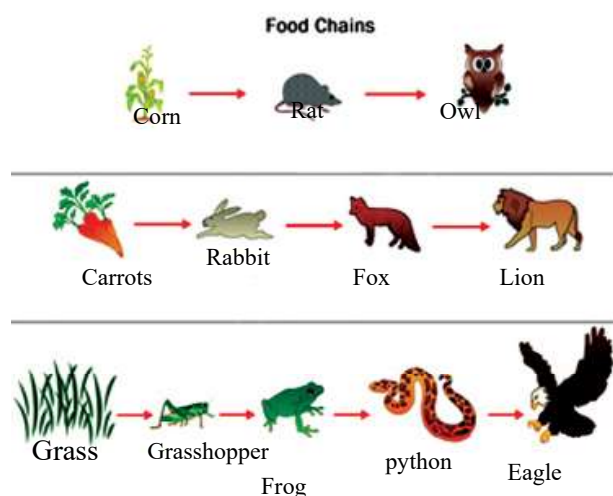


Figure 8.2: Food chains in a terrestrial ecosystem

Food web

In a natural ecosystem isolated food chains do not exist. The food chains are interconnected with each other to form food webs. The food web is an interconnected feeding relationships in an ecosystem. (Campbell, 10th edition, 2015)

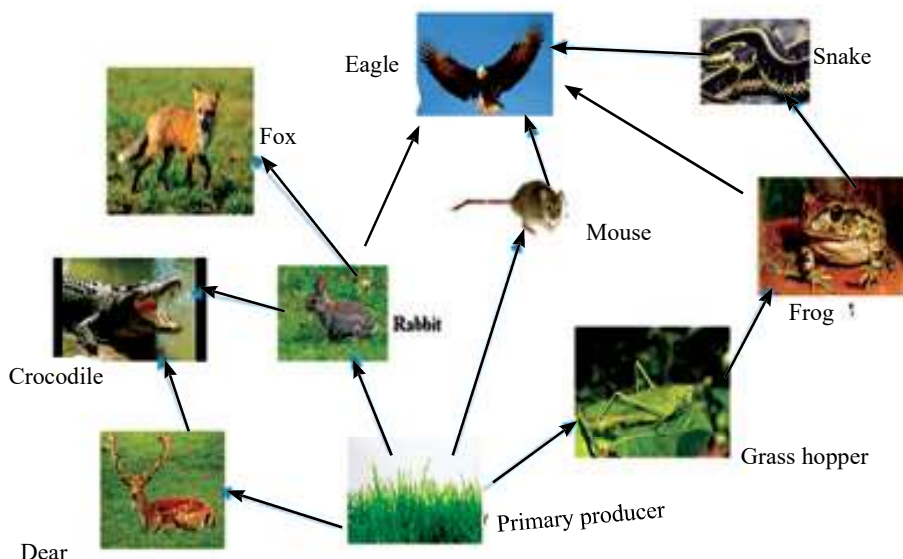


Figure 8.3 : Example for food web in a terrestrial ecosystem

Energy loss along food chains

The flow of energy along a food chain is always unidirectional. At each trophic level in the food chain, a considerable fraction (about 90 %) of the potential energy is lost as heat and respiration. As a result, organisms in each trophic level pass on lesser energy (about 10 %) to the next trophic level than they actually receive. This limits the number of trophic levels in any food chain to four or five. The most ecologically efficient food chain is the shortest one. Longer the food chain the lesser energy is available for top members. Because of this tapering off of available energy in the food chain a pyramid is formed that is known as an ecological pyramid. Shorter food chain has more energy available even at the highest trophic level than that of longer food chain.

Ecological pyramid:

The trophic structure of an ecosystem can be indicated by means of ecological pyramid. The higher the steps in the ecological pyramid the lower will be the number of individuals and the larger their size.

The concept of ecological pyramid was developed by Charles Elton; these pyramids are also known as Eltonian pyramids. The pyramids are a graphical representation which depicts the

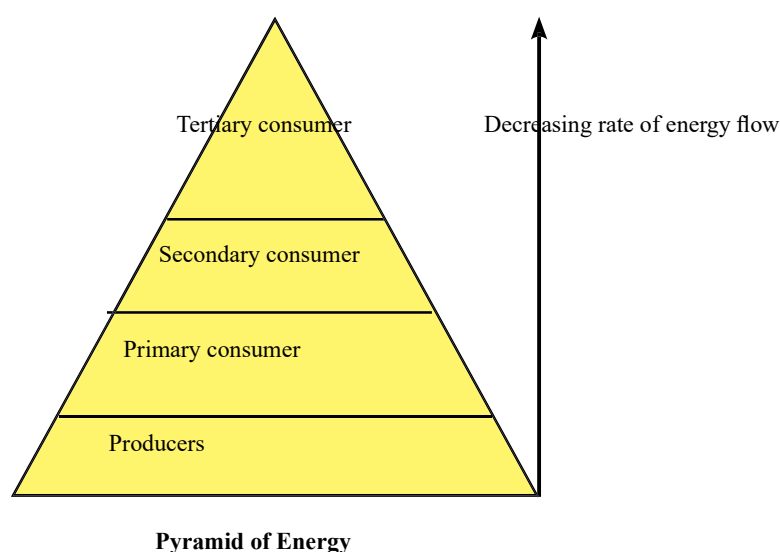
number of organisms, biomass and productivity at each trophic level. All ecological pyramids begin at the bottom with the primary producers and they proceed to various trophic levels such as herbivores consume plants, carnivores prey on herbivores and so on. The highest level is at the top of the food chain. There are three types of ecological pyramids:

- Pyramid of energy
- Pyramid of numbers
- Pyramid of biomass

Pyramid of Energy

Pyramid of energy represents the amount of energy at different trophic levels. The pyramid of energy or the energy pyramid describes the overall nature of the ecosystem. As there is considerable loss of energy during the flow of energy from organism to other, the energy pyramid always upright and vertical.

- This pyramid shows the flow of energy at different trophic levels.
- It depicts the energy is minimum as the highest trophic level and is maximum at the lowest trophic level.
- At each trophic level, there is successive loss of energy in the form of heat and respiration, etc.



Pyramid of Numbers

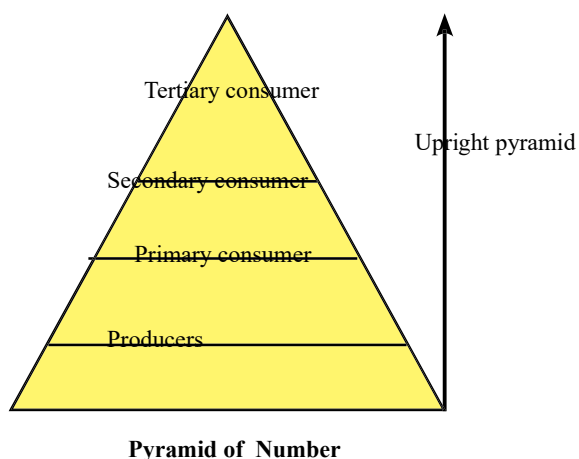
The pyramid of numbers depicts the relationship in terms of the number of producers, herbivores and the carnivores at their successive trophic levels. There is a decrease in the number of individuals from the lower to the higher trophic levels. The number pyramid varies from ecosystem to ecosystem. There are three types of pyramid of numbers:

- Upright pyramid of number
e.g. Producers → Herbivores → Carnivores

- Inverted pyramid of number.

e.g. Host $\longrightarrow$ Parasites

This type of pyramid number is found in the aquatic and grassland ecosystem, in these ecosystems there are numerous small autotrophs which support lesser herbivores which in turn support smaller number of carnivores and hence this pyramid is upright.



Pyramid of Biomass

The pyramid of biomass is more fundamental. In this pyramid there is a gradual decrease in the biomass from the producers to the higher trophic levels. As the fresh weight of biomass contains a larger amount of water, dry weight of the biomass represents the amount of energy available in the form of organic matter of an organism.

There are two types of pyramid of biomass, they are:

- Upright pyramid of biomass and -This occurs when the larger net biomass of producers support a smaller biomass of consumers.

Example: Forest ecosystem.

- Inverted pyramid of biomass.

e.g. In some aquatic ecosystems phytoplanktons supports larger primary consumers.

8.4: Upright pyramid of Biomass



Cycling of materials in an ecosystem

The materials available for living organisms in the ecosystem is limited. Therefore these materials should be recycled. The material stock in the ecosystem is decreased when the organisms such as animals, plants, etc utilize them. Therefore when the organism is died, they are decomposed by decomposers. Thus those materials will be available for the organisms and absorb them into their body, assimilate and the excess will be removed as waste materials to the environment.

What is a biome?

A biome is a large geographical area which is classified based on the predominant vegetation adapted to that particular environment. The climate and geography of a region determines what type of biome can exist in that region. Each biome consists of many ecosystems whose communities have adapted to the small differences in climate, topography and soil conditions within the biome. Major terrestrial biomes include tropical forest, savanna, desert, chaparral, temperate grassland, temperate broad leaf forest, northern coniferous forest and tundra.

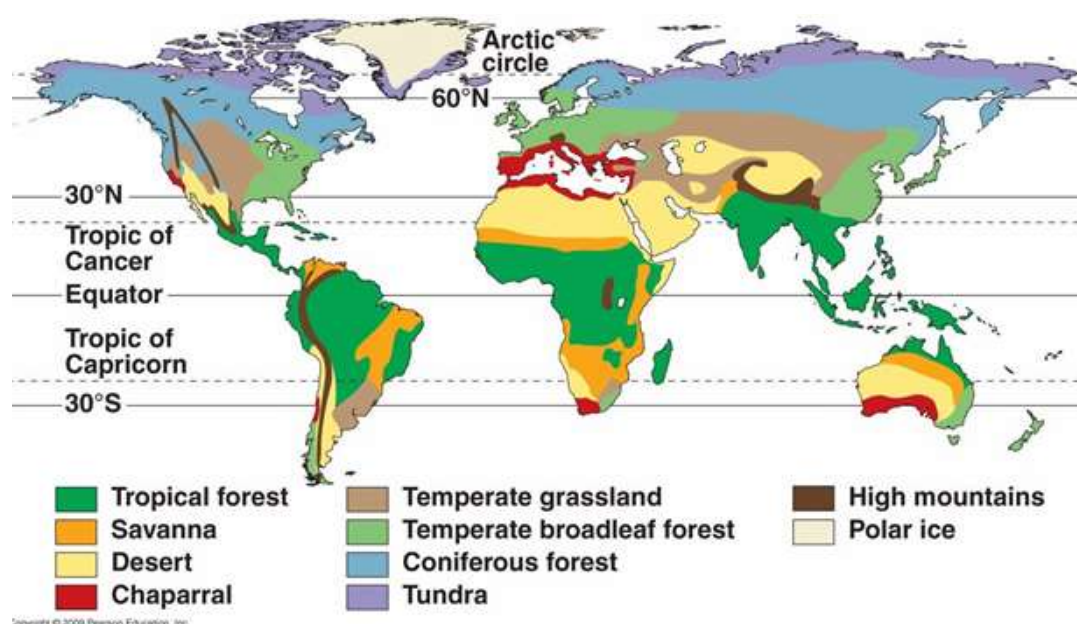


Figure 8.5: World distribution of different types of biomes

1. Tropical forests

This biome is distributed close to the equator and extends to the sub equatorial areas. Therefore both tropical rainforests and tropical dry forests are included.

The average annual rainfall in tropical rain forests is about 2000 -4000 mm and in tropical dry forests it is about 1500- 2000 mm. Seasonal rainfall with a dry season of 6-7 months is prominent in the dry forests and a fairly constant rainy season is prominent in the rainforests.

Tropical rainforests have an average temperature of 25 -29 °C, whereas in the tropical dry forests it may reach up to 33 °C.

In tropical rainforests an emergent layer, canopy and sub canopy layers are visible. These are followed by an understory layer which consists of shrubs, and large herbaceous plants. The forest floor consists of small herbaceous plants, mushrooms, and a thick layer of dried plant litter. Therefore the vegetation is arranged in several vertical layers and stratification is observed.

Evergreens are prominent in tropical rain forests whereas in dry forests leaves of deciduous species fall in the dry season. Epiphytes are common in this biome. However in dry forests they are less common. Shrubs with thorns as well as succulent plants often occur in dry forests. Tropical forests constitute the higher diversity of plants and animals out of all terrestrial biomes. It has many mammal species and 5-30 million species of arthropods some of which have not been fully described yet. Animals in this biome have adaptations to suit the environment.

Animals in the rainforests show year round activity. Camouflage is common among small animals. Large animals may have markings in their bodies. Birds often migrate to other regions during dry period.

Human involvement in agriculture and development related activities has led to destruction of these forests.



Figure 8.6: A tropical forest

2. Savanna

Savanna is spread close to the equator and subequatorial areas.

The average annual rainfall in this biome is around 300-500 mm. A prolonged dry season of approximately 8-9 months is prominent. The average temperature range is 24- 29 °C. However in subequatorial areas seasonal variations may occur.

Savannas are characterized by a landscape that supports scattered trees in a tall grass cover. During dry periods the dried grasses catch fire easily. Therefore grasses adapted to withstand drought, naturally grow in savannas. These grasses have a very good underground connectivity. The few herbs found among grass cover contain small leaves to reduce transpiration and thorns to be protected from herbivores.

Seasonal rains encourage a fresh growth of grasses to compensate the loss of biomass due to consumption by herbivores.

Many insects (mainly termites), lions, zebras are the common animals in this region. Many animals have effective locomotion for long distance migration and long range vision for



Figure 8.7: A Savanna

hunting. Human settlement from a very early time has been reported in these areas. Due to rearing of cattle and hunting of animals by humans the population of mammals have been decreased. Fires caused by human aid help to maintain the grass cover of this biome in one way but it may also have a negative impact by suppressing the growth of trees.

3. Desert

Desert is a temperate or tropical biome, commonly occurring near 30° north and south latitudes (e.g. Sahara), and in the center of continents (e.g. Gobi desert in north central Asia). The distribution of this biome is determined by the non availability of water such as in areas with annual average rainfall is less than 3000 mm per year. The temperature varies seasonally and daily and exceeds 50 °C in hot deserts while in cold deserts it drops below -30° C.

Deserts have more bare lands. If the vegetation is found they are sparse and widely scattered. Desert plants have several adaptations to withstand the high temperature and scarcity of water.

- Succulent plant body (e.g. Cacti and Euphorbs)
- Most plants have C4 pathway of photosynthesis
- Deep roots in shrubs
- Ability to tolerate heat and desiccation
- Reduced surface area of leaves
- Presence of spines/ thorns
- Presence of toxins in leaves
-

Nocturnal animal species are abundant in deserts. Water conservation is a prominent feature, with some species surviving solely on water obtained during the breaking down of carbohydrates in seeds. Animals include, snakes, lizards, beetles, ants, scorpions, rodents and birds.

Since these areas are converted into human settlements and agricultural lands, the natural biodiversity is reduced. Deep wells and well developed systems for transporting water has facilitated human settlement and agricultural lands.



Figure 8.8 : A Desert

4.Chaparral

Chaparral occurs in mid latitude coastal regions such as North America, Chile, Spain, Southern France and South Africa. Chaparral is best developed in Southern California.

Chaparrals receive an average annual rainfall of 300-500 mm. They experience rainy winters and dry summer. The average temperature is between 10 to 12 °C but can reach up to 40 °C.

The Chaparral biome is typically composed of dwarf forests and shrubs, and interspersed herbaceous vegetation which include grasses and herbs. Chaparral is highly prone to events of catastrophic wildfire. Therefore plants show the some of the following adaptations;

- Seed germination occurring only after a hot fire.
- fire resistant roots.
- Uses of food stored in the fire resistant roots for resprouting after a fire.
- Quick re -sprouting enables usage of nutrients released by the fire.
- Tough evergreen leaves in woody plants to survive in droughts.

Most of the native mammals in chaparrals are browsers. They include deers and goats. Chaparrals are rich in diversity of small mammals, and several species of amphibians, birds, reptiles and insects.

These areas have been reduced and disturbed due to human settlement, urbanization and agricultural conversions. Chaparral is vulnerable to fire caused by human activities.

5. Temperate grasslands

These grasslands occur under temperate climatic regimes that are intermediate to those that support forest and desert. Grasslands in North America are called prairie (they are often called

steppe in Eurasia and called pampas in Argentina).

Precipitation is highly seasonal with periodic drought. These grasslands have relatively dry winters and wet summers. The average rainfall is 300 to 1000 mm per year.

During winter seasons the average temperature falls below -10°C and in summer it is about 30°C .

Grasses are the dominant plant species in these grasslands. The prairie is often divided into three types according to height of the dominant vegetation—tall-grass, mixed-grass, and short-grass. The height of grasses vary from few centimeters to two meters in tall grass prairie. Fire and droughts are occurred in these grasslands but many plants growing in those areas have adaptations to cope with fire.

Large grazers such as horses and many types of burrowing mammals (prairie dogs in North America) are found in these areas.



The tall-grass prairie is now an endangered natural ecosystem, because it has been almost entirely converted to agriculture and farmlands.

Figure 8.9 : Temperate grassland

6. Temperate broadleaf forest

Distribution of temperate broadleaf forests are at mid latitudes in the northern hemisphere. The average annual precipitation is between 700 to 2000 mm. Significant amount of precipitation is received throughout the year in summer as rain and in winter as snow.

Average temperature during winter is zero while in summer it is up to 35°C . Summer is hot and humid and allow a favourable time to the growth of trees.

The dominant trees are mostly deciduous. A vertical layering (stratification) can be seen in temperate broadleaf forests. They are closed canopy layer, one or two strata of understory trees, shrub layer and herb layer. There are only few epiphytes can be found.

Many mammals can be seen and they hibernate during winter seasons while the birds species migrate to areas where climate is warmer.

Forests are disturbed due to logging, clearance for agriculture and for human settlements.

7. Northern coniferous forest

The largest biome on earth is the northern coniferous forest. It is extended as a broad band from upper band of Arctic tundra.

Precipitation: Annual average precipitation is around 300- 700 mm and a periodic drought is also common.

The temperature is -50°C in winter and 20°C in summer.

The northern forest is dominated by coniferous trees. e.g. Fir, Pine, spruce are common tree species.

Many conifers are conical in shape and this shape of trees prevents accumulation of snow, thus preventing the breaking down of branches. They have needle like leaves that prevent high transpiration.

The diversity of shrubs and herbs in these forests is less than that of temperate broad leaf forests because of the limitations in precipitation and warm temperature.

Commonly found mammals are brown bears, moose, and siberian tigers. During some period of time dominant plants can be killed by sudden attacks of insects.

These areas are not heavily populated by humans, but logging is a severe threat.



Figure 8.10 : Northern coniferous forest

8. Tundra

This biome covers expansive areas of the arctic region, amounting 20% of earth's land surface. Alpine tundra occurs at high altitudes on mountains, while arctic tundra occurs at high latitudes. Most tundras receive very small inputs of water as precipitation, but nevertheless their soil may be moist or wet because there is little evaporation in such cold climates, and deep drainage may be prevented by frozen soil.

The arctic tundra annually receives an average precipitation around 200 to 600 mm and the alpine tundra receives > 1000 mm precipitation respectively.

During the winter time temperature falls below -30°C and during the summer temperature it is always less than 10°C .

Mostly herbaceous plants can be seen that includes different types of grasses and forbs. Shrubs, mosses, lichens and trees also can be found in tundra. A layer of permafrost which is a permanently frozen layer of soil can be seen. This permafrost layer restrict the growth of the roots of plants.

Large grazing mammals such as Caribou and reindeer (migratory), musk oxen (resident) and predatory wolves, foxes, bears together with many species of migratory birds that nest during summer can be seen in this biome.

This area is sparsely colonized by humans, but heavily used for extraction of minerals and oils.



Figure 8.11 : The tundra

Ecosystems of Sri Lanka

Sri Lanka is in the north-equatorial tropical zone. Therefore it experiences a climate with high rainfall and temperature which permits the country to be blessed with a wide array of terrestrial and aquatic ecosystems. The ecosystems can be broadly classified into groups as follows.

1. Terrestrial ecosystems-
 - a. Forests-Lowland rain forests, Dry monsoon forests, Montane forests, Thorn scrubs
 - b. Grasslands- Savanna, patana
2. Inland wetland ecosystems- Rivers and streams, Reservoirs, Marshes and swamps, villus
3. Ecosystems is with coastal areas- Lagoons and estuaries, Mangroves, Coral reefs, Sea shore, Sand dunes, Sea grass beds, Salt marshes.

The distribution of major vegetation types may be considered as broadly linked to the local climate (mainly rain fall and temperature), topography, and the edaphic (soil) conditions. A broad classification of major ecosystems of Sri Lanka is given below.

Table 01: Classifications of eco systems in Sri Lanka

Climatic zone	Parameters T: temperature (°C) R: Rainfall (mm) Alt: Altitude (m)	Forests	Grasslands	Other
Arid zone	T 32-36 R < 1000 Alt < 300	Tropical thorn scrubs	Arid	Salt marsh Mangroves
Dry zone	T 28-32 R: 1000-1500 Alt < 500	Tropical dry mixed evergreen forests	Damana Talawa Savanna	Sea shore Dunes Coastal marine,
Intermediate zone	T 24-28 R 1250-2000 Alt 500-1500	Tropical moist evergreen forests, Tropical sub montane forests	Savanna Dry Patana	Reservoirs, rivers streams, and riverine Wetland Marshes Villus
Wet zone	T 16-28 R > 2000 Alt 300-1000 Alt 900-1500 Alt > 1500	Tropical lowland wet ever-green Tropical sub-montane Tropical upper montane	Talawa Dry Patana Wet Patana	

Source; National Atlas of Sri Lanka, 2nd Edition, Survey department of Sri Lanka

Tropical wet low land rainforests:

The tropical wet lowland rainforests are found below 900 m elevation in the south western quarter of the country. The mean annual temperature is about 28 °C, and the mean annual rainfalls varies between 2000 mm- 5000 mm in different locations without any dry period. The humidity is very high.

These forests are characterized by a tall canopy, sub canopy and a sparse shrub layer consisting mostly of the saplings of canopy trees. Sometimes an emergent layer is also seen above the canopy.

The vegetation is filled with a network of woody lianas that reach to the canopy layer.

A diverse population of epiphytic lichens, mosses, liverworts ferns, orchids are found while many fungi inhabit decaying woods and soil. High endemism of floral and faunal diversity can be seen in these forests. E.g. Sinharaja, Nakiyadeniya and Kanneliya are some examples for wet lowlands rainforests. These forests are disturbed due to expansion of tea cultivation, agriculture, industrial and residential development.

Some tree species found in wet lowland rain forests are;

- S: Hora, T: Ennai (*Dipterocarpus zeylanicus*)
- S: Naa, T: Nagai (*Mesua ferrea*)
- S: Hal, T: Kungiliyam pinai (*Vateria copallifera*)

Some animal species found in the wet lowland rainforests are Purple faced langur, Sri Lanka slender loris, Golden wet zone palm civet.



Figure 8.12 : Tropical wet lowland rain forests

Tropical montane forests

They occur at elevation beyond 1500 m above mean sea level. The average temperature is about

16 °C and rainfall is about 2000 mm without any dry period. The humidity is high as lowland rain forests.

These forests are characterized by a short canopy of about 13 m, with a dense shrub layer. There are trees with umbrella- shaped rounded crowns and twisted branches having leathery small leaves to accommodate the strong winds that prevail in the montane zone. High density of epiphytes cover the branches and stems of trees. Some plants found in these forests are;

- S: Keena, T: Pongu (*Callophyllum walkeri*)
- S: Walkurudu, T: Kaatu karuwa (*Cinnamomum ovalifolium*)
- S: Gal weralu, E: wild olive (*Elaeocarpus montanus*)

Some animals that inhabit montane rainforests of Sri Lanka are Sri Lanka Yellow eared bulbul, Sri Lanka highland Shrew, Sambar.

Montane forests are distributed in central hills including Knuckles range, Piduruthalagala, Hakgala. Montane forests were greatly affected due to tea cultivation.

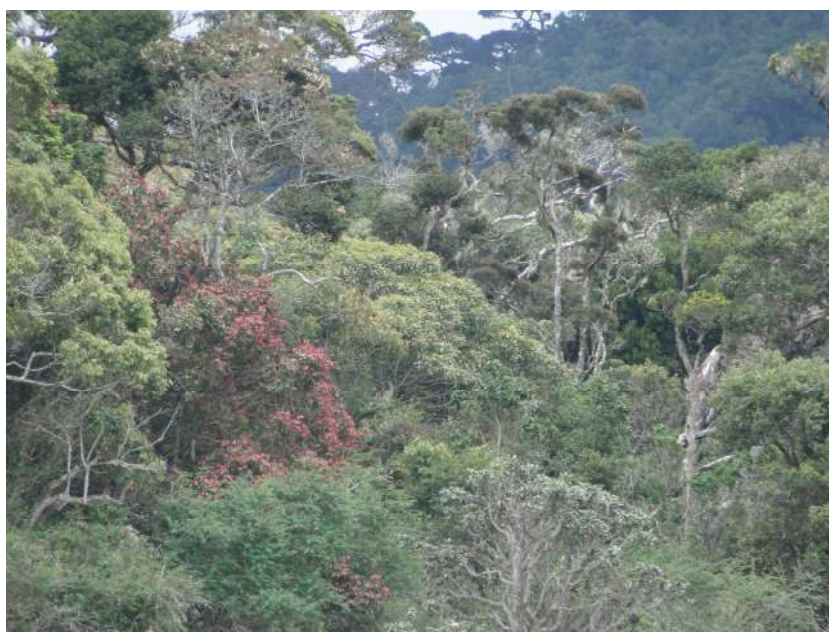


Figure 8.13 : Tropical montane forests

Tropical dry mixed evergreen forests/Dry monsoon forests

These forests have mean annual temperature of about 29 °C and rainfall around 1000 mm- 1500

mm., most of which falls during the northeast monsoon periods. There is a marked dry spell from May to August. Above forests are distributed in areas below 300m altitude.

These forests are characterized by having sparse canopy, a sub canopy and a well-developed shrub/herb layer. Some plant species naturally found in these forests are

- S: Weera T: Virai (*Drypetes sepiaraia*)
- S: Palu, T: Paalai (*Manilkara hexandra*)
- S: Kaluwara, T: Karun-kaali (*Diospyros ebenum*)

Some of the canopy trees are deciduous during the dry spell of the year. These forests harbor the largest elephant populations in Asia. The mammal fauna includes leopard, bear, deer. Most of the forests are protected as National parks such as Yala National park, Wilpattu National Park, Wasgamuwa National Park, Maduru-Oya national park, Ritigala Strict Nature reserve and many sanctuaries and forest reserves.

These forests are disturbed by ‘chena’ cultivation, human settlements, extraction of timber, poaching etc.



Figure 8.14: Tropical dry mixed evergreen forests

Tropical thorn scrubs

These are called “scrubs” than “forests” as large trees are sparse and the vegetation is mainly consists of thorny shrubs. They are found in arid lowlands. The mean annual temperature is around 31 °C and rainfall is below 1000 mm with a longer dry period.

The following plant species are commonly found ;

- S: Gini- andara , T:Vindattai (*Dichrostachys cineria*)
- S: Ranawara, T: Avaram poo (*Cassia auriculata*)
- S: Heeressa, T: Pirandai (*Cissus quadrangularis*)

Due to the dry conditions many fauna species cannot be found. However Deer, Leopard, elephants can be seen in these areas. They are distributed near Hambantota, Yala, Mannar, Puttalam areas and disturbed due to various development pressures.



Figure 8.15 : Tropical thorn scrubs

Savanna

Savannas have a thick grass cover and few scattered trees. These are common on hill slopes of the dry or intermediate zone. The thin layer of soil on hill slopes cannot support the growth of trees and the grass cover of savannas helps to bind soil particles and reduce erosion of soil.

Periodic fires are common to this ecosystem as the dried grass cover easily catch fire in dry period of the year. They have trees with fire resistant species.

Some plant species found in Savanna are trees are;

- S: Aralu, T: Kadukkay (*Terminalia chebula*)
- S: Nelli, T: Topu- nelli (*Phyllanthus emblica*)
- S: bulu , T: Adhan- koddai, tanti (*Terminalia bellirica*) and
- have grass species such as
- S: Mana, E: Citronella grass , Vasanai pullu (T) (*Cymbopogon nardus*)
- S: Iluk E: Cogon grass T:Tharpai pullu (*Imperata cylindrica*)

Savannas are seen in Bibile, Monaragala, Mahiyanganaya, Wellawaya areas. These are distributed

by periodic fires set by villagers for various reasons.

Patana

There are two types of 'Patana' found in Sri Lanka; wet patana grassland and dry patana grassland based on the rainfall pattern and soil of the site.

'Wet Patana' grass lands are found in areas above 1500 m from the sea level with mean annual rainfall of over 2000 mm. Temperature ranges from 5°C to 18° C. Mist, fog and frost are common in these areas and do not experience any dry period.

The grasses do not reach more than one meter height and are called tussock grasses which include the *Chrysopogon nodulibarbis* and *Arundinella villosa*. The animals include a large Sambar and Wild- boare populations and few leopards. Extensive "Wet Patana" are found only in Horton plains.

Dry patana grasslands are found in altitudes between 500 m to 1600 m. They receive a rainfall about 1400 mm to 2000 mm with a definite dry period. Temperature ranges from 18°C to 24° C. The vegetation is made up of grasses that grow up to 1-2 m height such as Pangiri mana (*Cymbopogon nardus*) and *Themeda*/ (Pini bara tana)/ *Themeda tremula*.

"Dry Patana" is common on hill tops in in Hantana, Gampola, Welimada, and Haputale .



Figure 8.16 : Dry Patana



Figure 8.17 : Wet Patana

Wetlands

Wetlands are simply habitats with permanent or temporary accumulation of water with associated plant and animals. According to Ramsar Convention, wetlands are defined as areas of marsh, fen, peatland or water, whether natural or artificial, permanent or temporary with water that is static or flowing, fresh, brackish or salt, including areas of marine water the depth of which at low tide not exceed six metres. The wetlands of Sri Lanka, which fit into the Ramsar definition, can be divided into three broad categories:

- Inland fresh water wetlands (e.g. rivers, stream, marshes, swamp forests and villus)
- Coastal wetlands (e.g. lagoons, estuaries, mangroves, sea grass beds, salt marshes and coral reefs)
- Man-made wetlands (e.g. tanks, reservoirs, rice fields and salterns)

Rivers and streams

Sri Lanka has an extensive network of rivers, which drains a total of 103 distinct natural river basins. The river basins originating in the wet highlands are perennial, while many of those in dry zone are seasonal. There is hardly any vegetation to be found in running water.



Figure 8.18 : Streams

Marshes and swamp forests

Inland freshwater marshlands are low lying areas which receive water through surface runoff, ground water seepage or flood water from rivers. These contain peat (partially decomposed organic matter), and water logged sticky clay soil. Many water birds, amphibians and fish species inhabit these areas. Plants that have adapted to grow in shallow stagnant water such as habarala (S) / Semaikilangu (T)- *Colocasia* species, Kekatiya (S)- *Aponogeton* spp, Reeds (E)/ Pan (S) species are abundant in inland marshlands.

Freshwater swamp forests are not common in Sri Lanka. It is a place where forest vegetation is inundated for a short period of time in the year as seen in the Waturana swamp at Bulathsinhala located in the 'kalu ganga basin'.

Villus

Generally villus are the flood plains of the reservoirs. Villu grasslands possess a special link with

the wildlife in the area, especially elephants and bird populations. The vegetation is dominant with grasses (members of Poaceae) and sedges (members of Cyperaceae). Villu grasslands are located in areas such as Mahawelli flood plains, Wilpattu National park etc.



Figure 8.19 : Villu

Lagoons and Estuaries

Lagoons are coastal wetlands generally separated from the sea by a sand barrier. Eg: Negambo and Bundala lagoons. Estuaries are formed in places where rivers enter the sea and does not contain a sand barrier separating it from seas (e.g. Maduganga/Benthota). The daily tidal fluctuation is a characteristic of these places.

Mangroves

Mangroves ecosystem is an intertidal vegetation that covers fringes of the lagoons and estuaries. They share characteristics usually of saline/brackish water, loose soil and anoxic conditions. The vegetation is also exposed to intense sunlight.

Mangrove plants could be categorized into two groups; true mangroves that occur towards the boundary between sea and land and mangrove associates that occur more towards inland

Common true mangrove species are;

- S: kadol, T: kandal. (*Rhizophora* spp, *Bruguiera* spp,)
- S: Mas athu gas T: kannamaram (*Avicennia marina*)

Common mangrove associates are:

- S: Karan Koku, E: Golden leather fern (*Acrostichum aureu*) S:Katu- ikili E: Holly mangrove (*Acanthus ilicifolius*)

In order to protect from sunlight leaves of mangrove have a thick cuticle to reflect sunrays and/or cut off excess radiation. Some have salt glands to secrete excess salts that are absorbed by the plants. There are special roots to intake atmospheric oxygen. Seeds of some mangroves germinate while attached to the mother plant (vivipary). This help the seed to establish successfully soon after falling from the tree.